# Predictive vs Generative

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

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WORKBOOK · 1 OF 21

## Recall and choose your starting point

> Record your starting explanation for Predictive vs Generative before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

### Image and Voice Pipelines

Without opening the old lesson, explain one key idea from Image and Voice Pipelines. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

### Feedback Loops

Without opening the old lesson, explain one key idea from Feedback Loops. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

### Similarity by Selected Traits

Without opening the old lesson, explain one key idea from Similarity by Selected Traits. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

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WORKBOOK · 2 OF 21

## Practise with fading help

> Complete the supported Predictive vs Generative trace and record the first step you can explain without help.

Use W2: Compute the distribution for **Generative** and **predictive** objectives. Record the answer once; the lesson and workbook refer to the same attempt.

| Given inputs | Procedure I chose | Middle steps | Result and check |
|---|---|---|---|
| | | | |

**The step I completed without help, and the hint I needed**

WORKBOOK · 3 OF 21

## Individual reasoning check

> Check the proposed Predictive vs Generative explanation against the supplied facts before discussing it.

Use the worked case for **Generative** and **predictive** objectives. Proposed explanation: "**Generative** models do not predict". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

**My explanation before partner discussion**

WORKBOOK · 4 OF 21

## Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

**Independent case: Analyse a new corpus independently**

For prefix The signal is, counts are high 2, low 5, unknown 3. Compute all probabilities and the most frequent continuation. A current measurement reports high.

State what the most frequent continuation does and does not establish.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

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WORKBOOK · 5 OF 21

## Your learning target

> Connect the Predictive vs Generative target to a trace and a justified conclusion.

A tool retrieves a book description, predicts a genre, then writes a new blurb. Identify each operation and give one distinct failure to check at each stage.

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WORKBOOK · 6 OF 21

## Solve the given case

> Solve the supplied Predictive vs Generative case and show the deciding evidence.

A labels genre; B writes a new blurb; C computes 24/6.

Distinguish operations.

Show your trace, calculation or **evidence**

**Write or draw your reasoning and evidence.**

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WORKBOOK · 7 OF 21

## Check a changed case

Apply the procedure to the changed case and record why the conclusion follows.

One model labels a review positive or negative; another writes a new review. Which classifies and which generates? Must generated text be true?

My independent answer and reason

**Write or draw your reasoning and evidence.**

**Week 27 Test and reflect**

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WORKBOOK · 8 OF 21

## Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

| Test record | My evidence |
| --- | --- |
| Changed input or condition | |
| Expected answer or fallback | |
| Observed result | |
| Why the result occurred | |

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## Project checkpoint

> Justify what Predictive vs Generative adds to Transparent Book Recommender and keep the evidence.

Keep scoring **predictive** and explanations templated or carefully generated and verified.

The project change and **evidence** I will keep

**Write or draw your reasoning and evidence.**

WORKBOOK · 10 OF 21

## Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

## Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

A tool retrieves a book description, predicts a genre, then writes a new blurb. Identify each operation and give one distinct failure to check at each stage.

**My independently reasoned answer**

Record support used: none, hint or teacher explanation.

WORKBOOK · 12 OF 21

## W1 Identify the objective

> Identify the objective. Apply the supplied Predictive vs Generative facts and keep your reasoning.

Classify these components by target: label a message request/report; continue a prefix with a sequence; retrieve a stored sensor record. Explain why the application may combine them.

**My reasoning and evidence**

WORKBOOK · 13 OF 21

## W2 Compute the distribution

> Compute the distribution. Apply the supplied Predictive vs Generative facts and keep your reasoning.

For The marker is, calculate the three empirical probabilities, their sum and their meaning.

**My reasoning and evidence**

WORKBOOK · 14 OF 21

## W3 Follow the prefix

> Follow the prefix. Apply the supplied Predictive vs Generative facts and keep your reasoning.

After red is selected, the expanded-prefix table gives end 0.8 and today 0.2. Explain why these values cannot be substituted for the red/blue/green probabilities.

**My reasoning and evidence**

WORKBOOK · 15 OF 21

## W4 Analyse a new corpus independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Analyse a new corpus independently. Apply the supplied Predictive vs Generative facts and keep your reasoning.

For prefix The signal is, counts are high 2, low 5, unknown 3. Compute all probabilities and the most frequent continuation. A current measurement reports high.

State what the most frequent continuation does and does not establish.

**My reasoning and evidence**

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WORKBOOK · 16 OF 21

## W5 Design a useful evaluation

> Design a useful evaluation. Apply the supplied Predictive vs Generative facts and keep your reasoning.

A system reports current room status using retrieved measurements and generated sentences. Name checks for **evidence**, response quality and permission.

**My reasoning and evidence**

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WORKBOOK · 17 OF 21

## Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

| Evidence | My record |
| --- | --- |
| Worked alone / hint / model | |
| First step I repaired and why | |
| New case checked later; date and result | |
| What I can now explain without notes | |
| One remaining question | |

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

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**WORKBOOK · 18 OF 21**

## Pictorial case practice: Predictive vs Generative

> A fictional toy generator predicts the next word after “The box is”. Explain the labelled step before checking the changed condition.

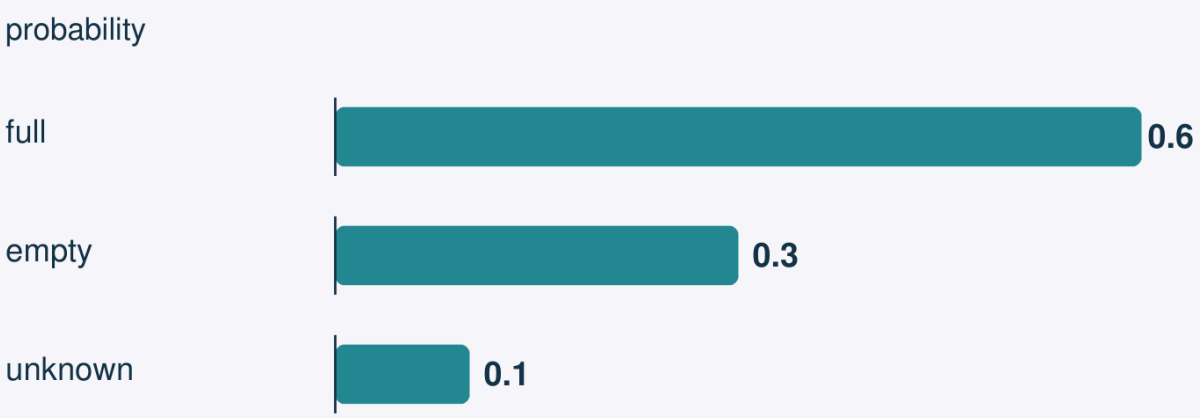
A fictional toy generator predicts the next word after “The box is”.

Observed **training** continuations: full 6, empty 3, unknown 1.

Use counts divided by total 10. The current physical box is separately checked and found empty.

AI ACADEMY / AI-2/27

## Toy next-word distribution




Bar length shows the supplied value; the marked baseline is zero.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-2/27

## Predictive vs Generative

**Start with the supplied facts**

Follow each arrow. Ask which fact allows the next step.



### 1. Count table

Continuation counts are 6,3,1; total 10.



### 2. Probabilities

full=0.6; empty=0.3; unknown=0.1.



### 3. Greedy continuation

Largest probability chooses full for this toy prefix.



### 4. Check reality

Current observation says empty; frequent wording is not a present inventory reading.

**CHANGE ONE THING**

Change the training table to full 2, empty 7, unknown 1.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Does matching the physical box in the changed table prove the generator measures boxes?

**My picture-based explanation and changed-case prediction**

### Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Use approved book facts for recommender explanations; generated fluency is not a replacement for catalogue **evidence**.

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WORKBOOK · 19 OF 21

## Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Transparent Book Recommender record.

Project: Transparent Book Recommender

Current milestone: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.

Use approved book facts for recommender explanations; generated fluency is not a replacement for catalogue **evidence**.

| Evidence to keep | My record |
| --- | --- |
| This week's input and deciding step | |
| Prediction and actual result (or labelled paper trace) | |

| Evidence to keep | My record |
| --- | --- |
| One change since the previous checkpoint and why | |
| One error or limit and the next test | |

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

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WORKBOOK · 20 OF 21

## Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional toy generator predicts the next word after "The box is".

### Practice 1: explain the deciding evidence

Which labelled step matches this detail? Current **observation** says empty; frequent wording is not a present inventory reading.

1. Probabilities

2. Count table

3. Greedy continuation

4. Check reality

**My selected conclusion, deciding fact and explanation**

### Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

### Feedback · compare after trying

After your attempt, compare with the supported explanation: Check reality Current **observation** says empty; frequent wording is not a present inventory reading.

The table predicts a continuation in a text exercise, not the contents of the current box. Generating the most frequent token is a model-defined decision. A factual inventory statement needs current **evidence**.

The numerical table is intentionally small; real **generative** systems condition on richer context and do not simply consult these three counts.

**Practice 2: explain the deciding evidence**

### Changed case · predict first

Changed-case practice: Change the **training** table to full 2, empty 7, unknown 1. Which conclusion is supported by the supplied facts?

1. Greedy **output** becomes empty with probability 0.7.

2. This one example guarantees correct results for every future **input**.

3. We can decide the result without examining the changed **input** or the procedure.

**My selected conclusion, deciding fact and explanation**

### Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

### Feedback · compare after trying

After your attempt, compare with the supported explanation: Greedy **output** becomes empty with probability 0.7. The continuation distribution changed; the independently observed box has not changed.

### Repair a missing building block

Review Image and Voice Pipelines (ai-2/26). Goal: Explain the **mechanism** and terminology of speech pipelines and multimodal **evidence**.

Short practice: Which stage first fails, and why should the faulty transcript not be silently overwritten?. Return to this week and explain what you changed.

Review Feedback Loops (ai-2/24). Goal: Implement a deterministic recommender. Short practice: What extra field would reveal why comparing raw clicks is misleading?. Return to this week and explain what you changed.

Review Similarity by Selected Traits (ai-2/21). Goal: Represent items with numeric features. Short practice: Explain why B ceases to be a closest option without any candidate data changing..

Return to this week and explain what you changed.

### Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

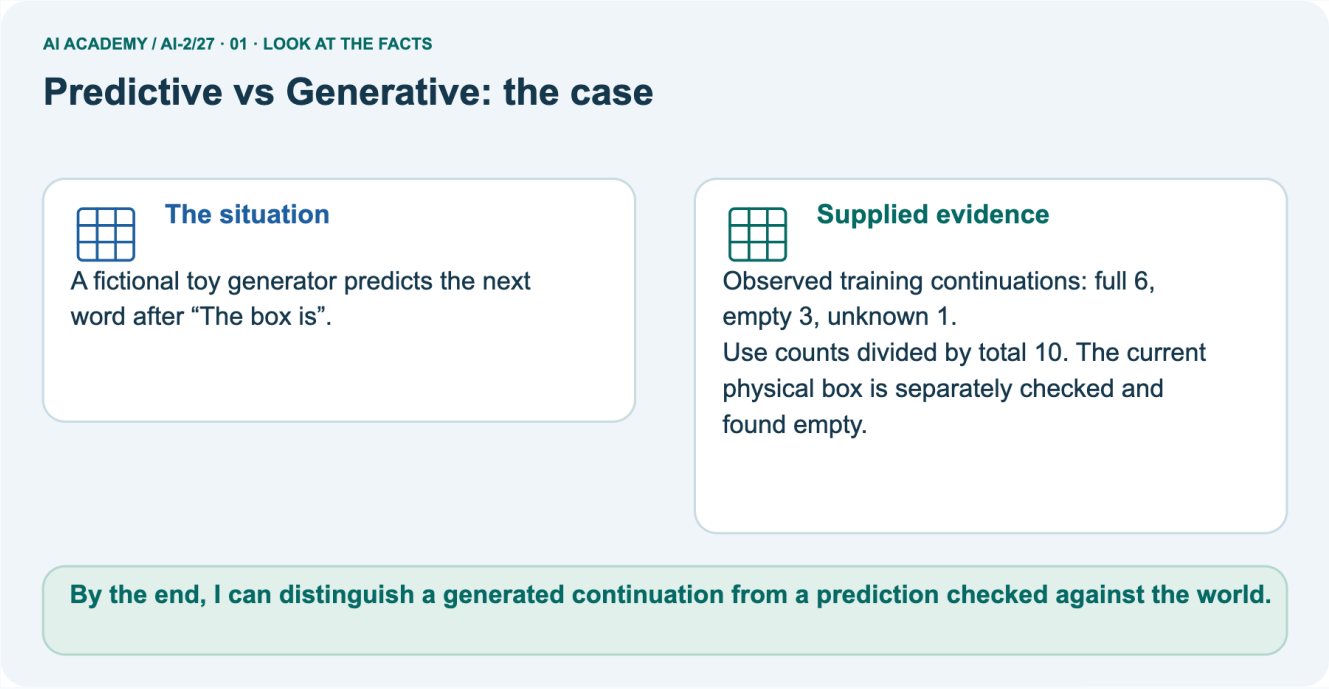
Drafts stay in this browser when storage is available; clear drafts on a shared device.

**What I repaired, the evidence I kept and the next test**

## Visual concept journey

> Explain each Predictive vs Generative picture from its labels, then record a relevant limit.

### 01. Predictive vs Generative: the case




Follow the labelled picture with the matching explanation. Open the vector diagram

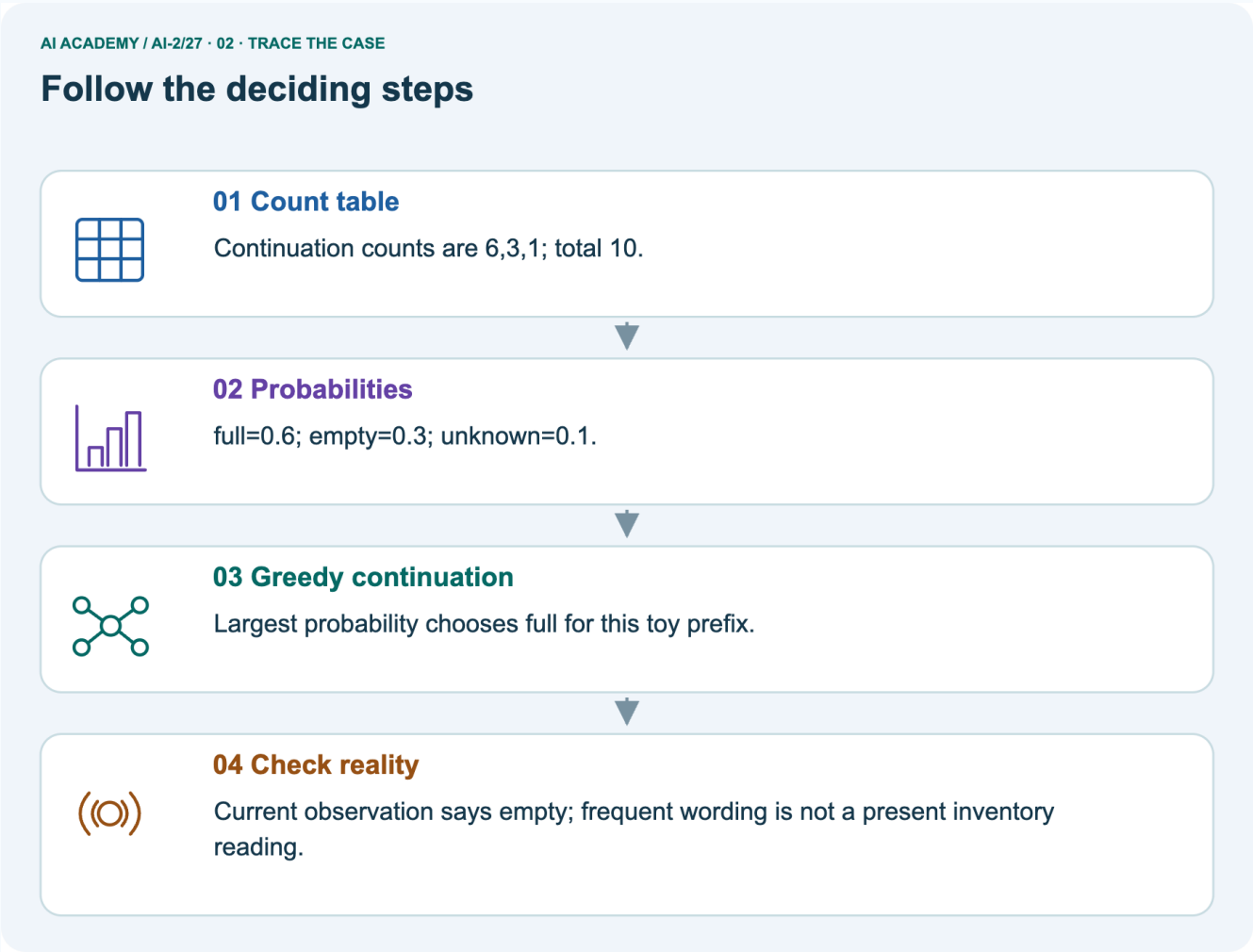
A fictional toy generator predicts the next word after “The box is”. Supplied facts: Observed **training** continuations: full 6, empty 3, unknown 1. Use counts divided by total 10.

The current physical box is separately checked and found empty.

> Point to the given fact that matters most. Explain why it matters.

### My picture explanation, deciding evidence and one limit

## 02. Follow the deciding steps



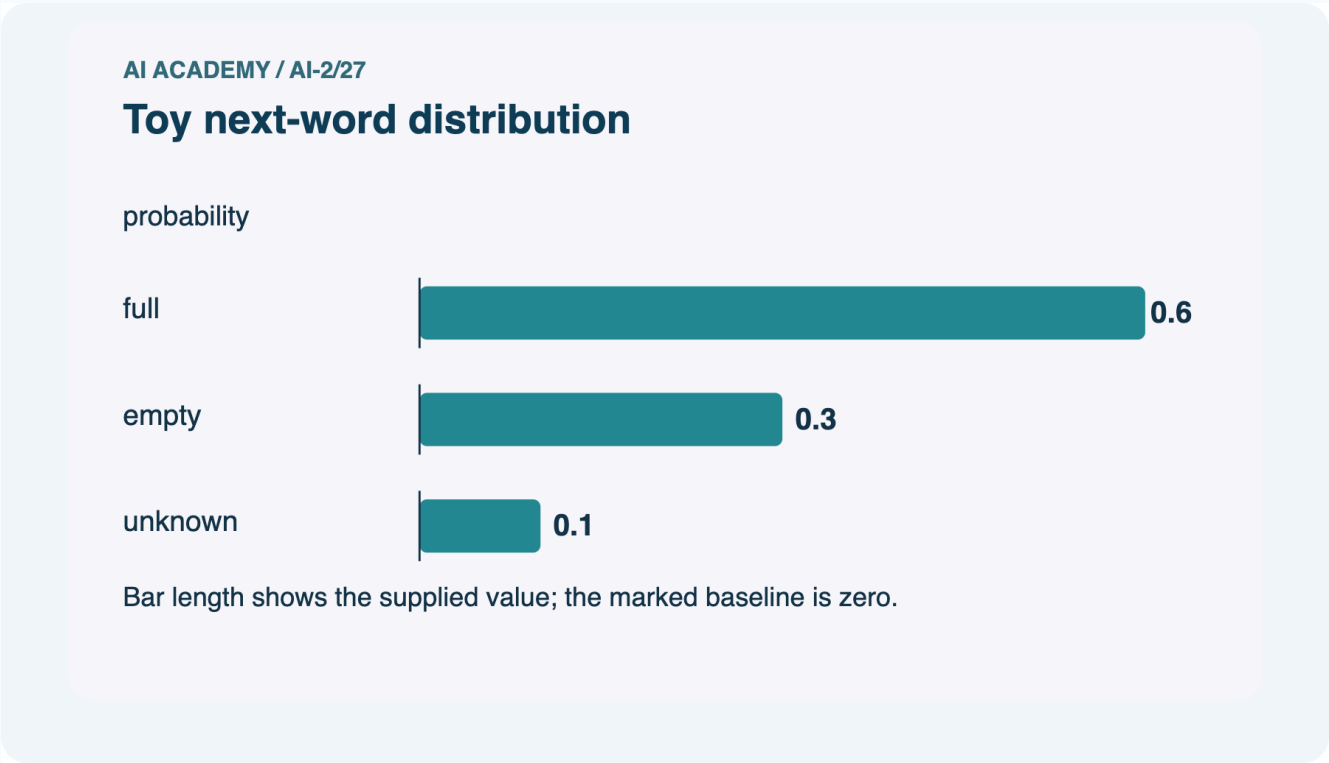

Follow the labelled picture with the matching explanation. Open the vector diagram

Count table: Continuation counts are 6,3,1; total 10. → Probabilities: full=0.6; empty=0.3; unknown=0.1. → Greedy continuation: Largest probability chooses full for this toy prefix. → Check reality: Current **observation** says empty; frequent wording is not a present inventory reading.

Point to each numbered stage and explain which recorded **input** it uses.

**My picture explanation, deciding evidence and one limit**

### 03. Toy next-word distribution



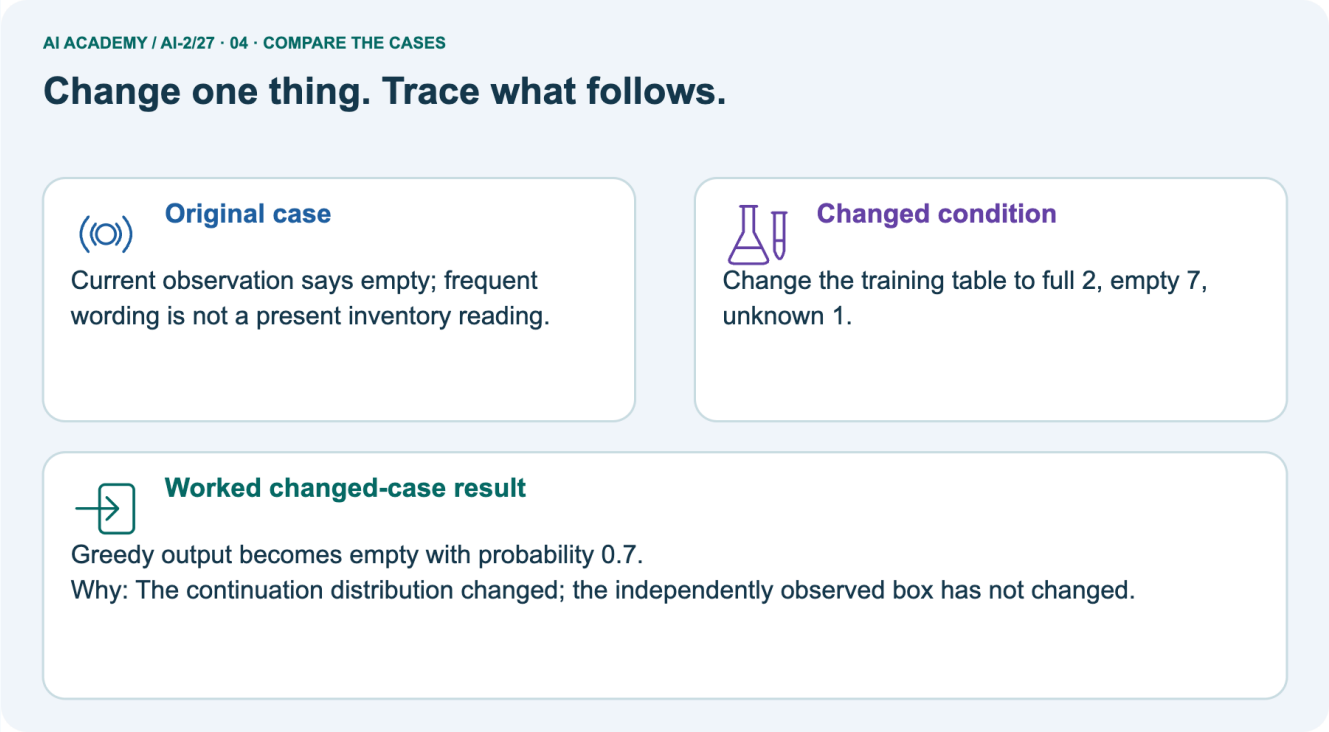

Follow the labelled picture with the matching explanation. Open the vector diagram

Toy next-word distribution. Units: probability. full: 0.6. empty: 0.3. unknown: 0.1.

Read the labels and values. Explain the relationship the model shows.

**My picture explanation, deciding evidence and one limit**

## 04. Change one thing. Trace what follows.




Follow the labelled picture with the matching explanation. Open the vector diagram

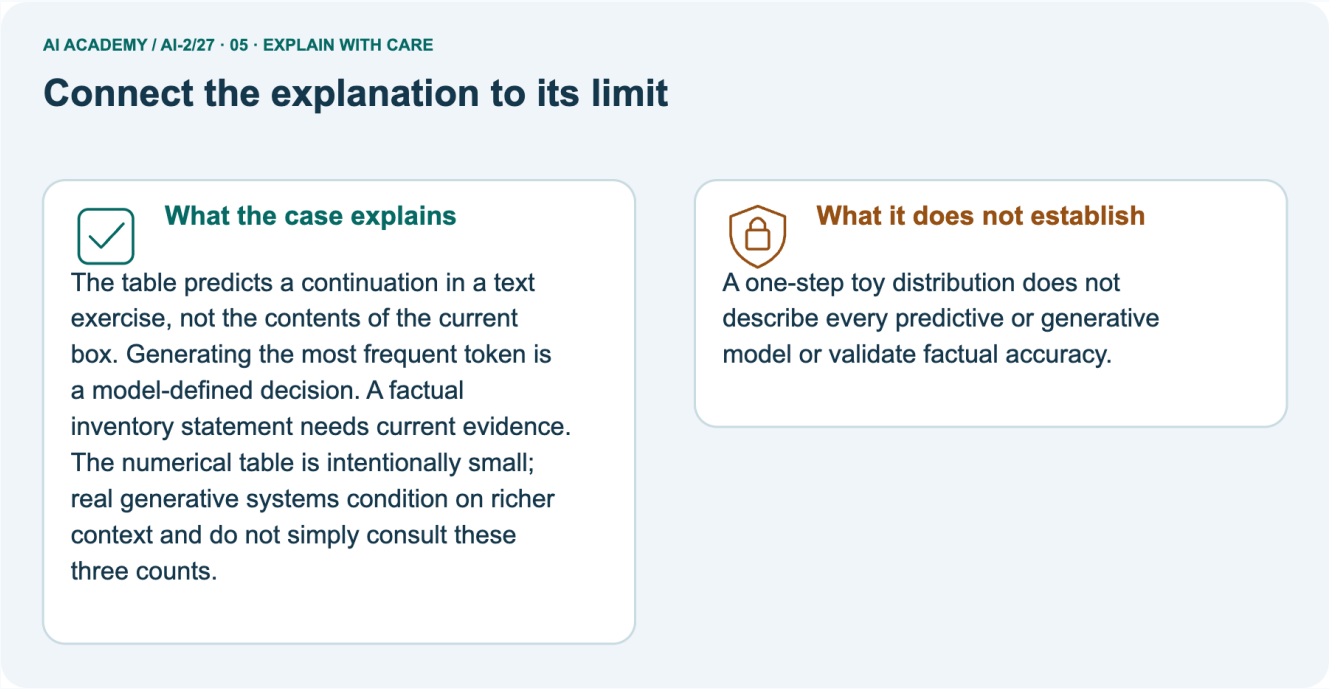
Original case: Current **observation** says empty; frequent wording is not a present inventory reading. Changed condition: Change the **training** table to full 2, empty 7, unknown 1.

Expected result: Greedy **output** becomes empty with probability 0.7. Reason: The continuation distribution changed; the independently observed box has not changed.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

**My picture explanation, deciding evidence and one limit**

## 05. Connect the explanation to its limit




Follow the labelled picture with the matching explanation. Open the vector diagram

The table predicts a continuation in a text exercise, not the contents of the current box. Generating the most frequent token is a model-defined decision. A factual inventory statement needs current **evidence**.

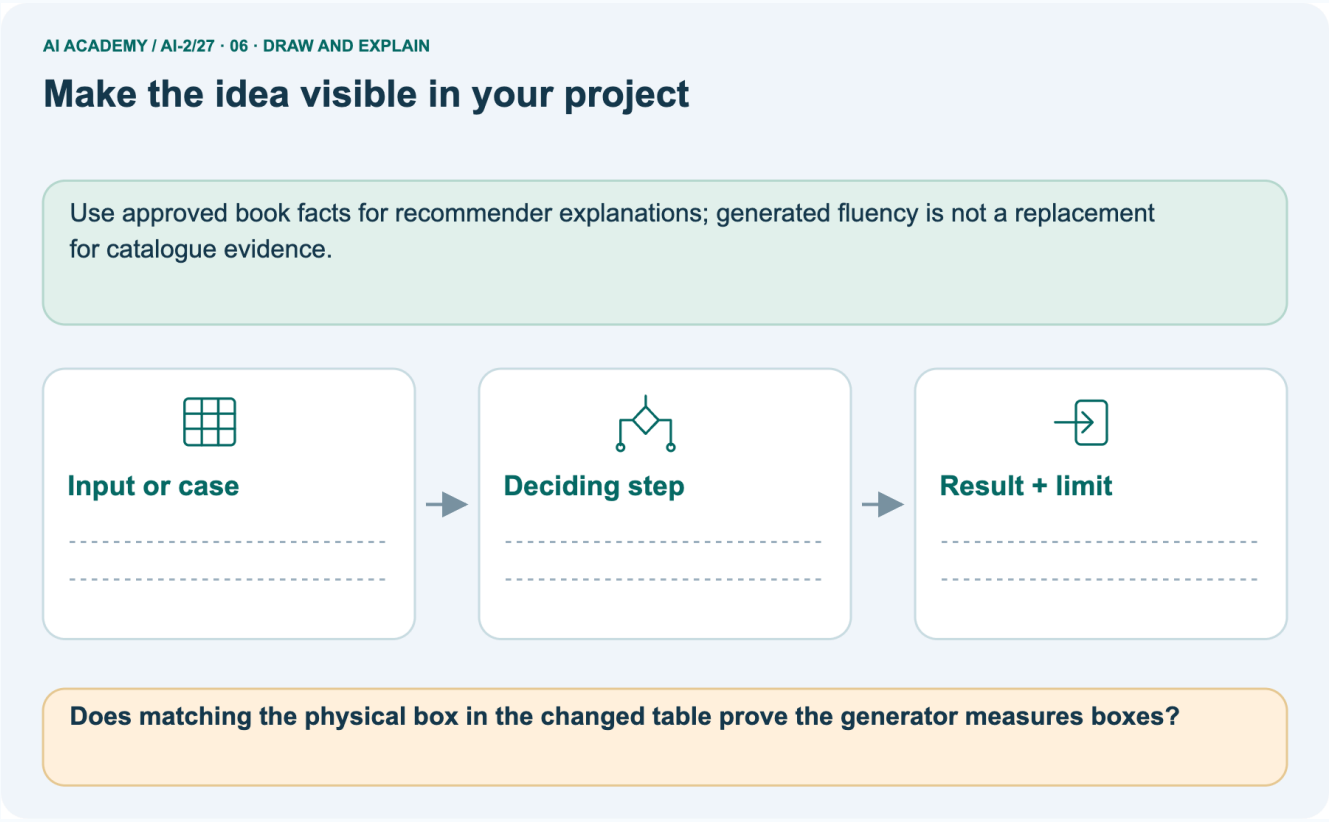
The numerical table is intentionally small; real **generative** systems condition on richer context and do not simply consult these three counts.

Limit: A one-step toy distribution does not describe every **predictive** or **generative** model or validate factual accuracy.

Explain the case, then point to a limit on the claim.

**My picture explanation, deciding evidence and one limit**

## 06. Make the idea visible in your project




Follow the labelled picture with the matching explanation. Open the vector diagram

Use approved book facts for recommender explanations; generated fluency is not a replacement for catalogue **evidence**. Draw your **input**, deciding step, result and limit.

Supported question: Does matching the physical box in the changed table prove the generator measures boxes?

Draw your project trace and justify the deciding step using **evidence**.

### My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.